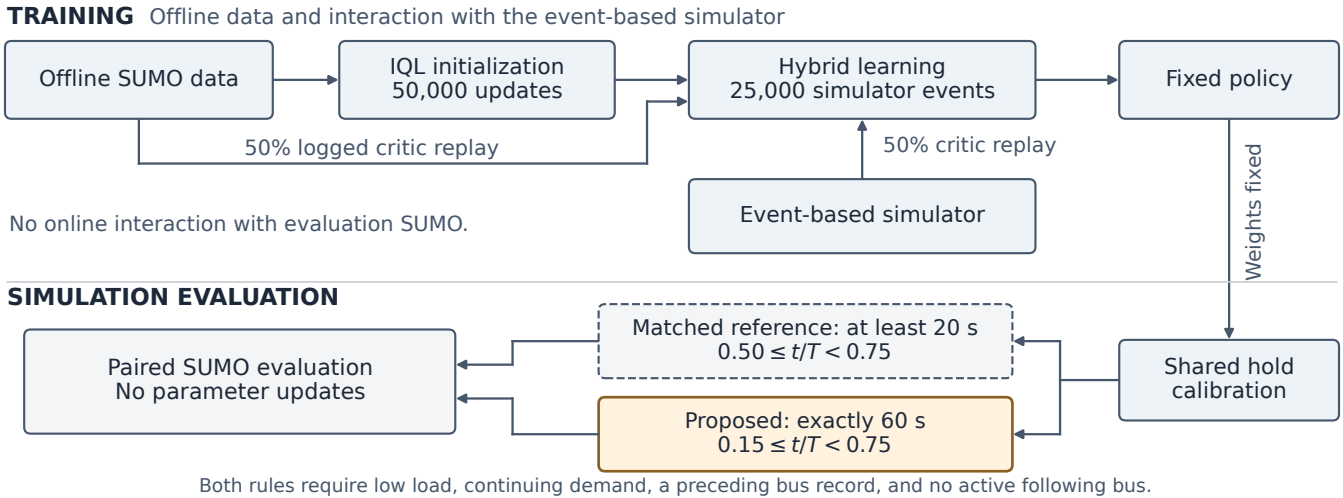
TRAINING Offline data and interaction with the event-based simulator
Offline SUMO data
IQL initialization
50,000 updates
Hybrid learning
25,000 simulator events
Fixed policy
50% logged critic replay
50% critic replay
Event-based simulator
Weights fixed
No online interaction with evaluation SUMO.
SIMULATION EVALUATION
Matched reference: at least 20 s
0.50 ≤ t/T < 0.75
Paired SUMO evaluation
No parameter updates
Shared hold
calibration
Proposed: exactly 60 s
0.15 ≤ t/T < 0.75
Both rules require low load, continuing demand, a preceding bus record, and no active following bus.